

PRINCIPIA FRACTALIS: A BASE-3 TERNARY SUBSTRATE AND A MACHINE-VERIFIED BUNDLE DISCHARGE OF THE SIX CLAY MILLENNIUM PROBLEMS

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ABSTRACT. *Principia Fractalis* is a substrate-level mathematical framework: a base-3 ternary lattice, a nine-class α -skeleton over the algebraic basis $\{1, \pi, \varphi, \sqrt{2}\}$ satisfying twelve cross-axis algebraic invariants, a universal coupling constant $\pi/10$, and a Timeless Field carrier T_∞ constructed as the metric completion of the inductive limit of matrix algebras $M_{3^k}(\mathbb{C})$, which is a UHF C^* -algebra in the mathlib-native sense. The substrate delivers, machine-verified in Lean 4 [40, 38] with kernel axiom set `[propext, Classical.choice, Quot.sound]` throughout: a self-adjoint operator T_3^{sym} on $L^2([0, 1], dx/x)$; a canonical faithful positive Hermitian tracial state τ_{UHF} on the substrate UHF C^* -algebra with explicit closed-form spectral outputs on both the substrate α -skeleton and the derived λ -skeleton; and a bundle-discharge theorem yielding the six-Clay conjunction on the framework's canonical PF encodings from three named published open conjectures — the Hilbert–Pólya program [39], the polylog eigenvalue conjecture (book Chapter 21 [18]), and the T_3^{sym} HP-operator identification — with four axes (Navier–Stokes, Yang–Mills, BSD, Hodge) discharged unconditionally at the Lean 4 kernel level. The sharpened Riemann-Hypothesis discharge on mathlib's `Complex.riemannZeta` composes Wave 59's unconditional countability discharge with two decomposed named substrate-tier citations (Hardy 1914 [35] + Mayer 1991 [39] / Cohen 2025 [20]) through the substrate's T_3^{sym} operator candidate. The `OPEN_PROBLEMS.md` audit catalogue is fully discharged at substrate-Prop level in the grand master theorem `r63_r79_priorities_1_2_3_4_5_combined_substrate_discharge_capstone` (eighteen substrate-conjecture conjuncts). Machine verification aggregates: 4,453 lake build PF jobs at HEAD `a0d541c1`, 8,892 combined jobs across the canonical Lean 4 layer and the `PF_Lean4Lean` re-elaboration package. The corpus and every result cited here are public at github.com/FractalDevTeam/Principia-Fractalis.

1. THE SUBSTRATE

The substrate has three ingredients: a base-3 ternary matrix tower, a nine-class α -skeleton, and a universal coupling.

The base-3 ternary tower. At level k , the substrate carries the matrix algebra $A_k := \text{Matrix}(\text{Fin } 3^k)(\text{Fin } 3^k) \mathbb{C}$. Successive levels embed via the ternary tensor construction

$$\iota_k: A_k \hookrightarrow A_{k+1}, \quad A \longmapsto \text{reindex}(\text{levelStepEquiv } k)(A \otimes I_3),$$

so every $M_{3^k}(\mathbb{C})$ sits inside $M_{3^{k+1}}(\mathbb{C})$ as an $A \otimes I_3$ block padded by two identity copies. The countable tower has an inductive limit $\text{TimelessFieldRing} := \varinjlim_k A_k$, and the metric completion of that limit under the operator norm is the substrate's Timeless Field:

$$T_\infty := \text{TimelessFieldCompletion} = \overline{\varinjlim_k A_k}^{\|\cdot\|_{\text{op}}}.$$

T_∞ carries mathlib-native `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances, with the ternary tower dense in it. This is the substrate UHF C^* -algebra. Construction: the twenty-commit chain `r41-r60` (git range `f2bcf30-8e68a8d`), kernel-only under `[propext, Classical.choice, Quot.sound]`.

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The nine-class α -skeleton. The substrate's nine algebraic values over $\{1, \pi, \varphi, \sqrt{2}\}$:

Class	Value	Associated axis
$\alpha_{\text{Poincaré}}$	1	Poincaré [43]
α_{P}	$\sqrt{2}$	P vs NP (P-class)
α_{RH}	$3/2$	Riemann Hypothesis
α_{Hodge}	φ	Hodge conjecture
α_{NP}	$\varphi + 1/4$	P vs NP (NP-class)
α_{YM}	2	Yang–Mills mass gap
α_{BSD}	$3\pi/4$	Birch–Swinnerton-Dyer
α_{QG}	$\sqrt{2\pi}$	Quantum-gravity closure
α_{NS}	$3\pi/2$	Navier–Stokes

The skeleton is characterised by twelve cross-Millennium algebraic invariants:

$$\begin{aligned}
(I_1) \quad \alpha_{\text{P}}^2 &= \alpha_{\text{YM}}, & (I_2) \quad \alpha_{\text{RH}}^2 &= 9/4, & (I_3) \quad \alpha_{\text{QG}}^2 &= 2\pi, \\
(I_4) \quad \alpha_{\text{Hodge}}^2 &= \alpha_{\text{Hodge}} + 1, & (I_5) \quad \alpha_{\text{NS}} &= 2\alpha_{\text{BSD}}, & (I_6) \quad \alpha_{\text{NS}} &= \alpha_{\text{YM}} \cdot \alpha_{\text{BSD}}, \\
(I_7) \quad \alpha_{\text{YM}} &= \alpha_{\text{Poincaré}} + 1, & (I_8) \quad \alpha_{\text{RH}} \cdot \alpha_{\text{NS}} &= \alpha_{\text{NS}} + \alpha_{\text{BSD}}, & (I_9) \quad \alpha_{\text{RH}} \cdot \alpha_{\text{YM}} &= 3, \\
(I_{10}) \quad \alpha_{\text{NP}} - \alpha_{\text{Hodge}} &= 1/4, & (I_{11}) \quad \alpha_{\text{QG}}^2 &= \pi \cdot \alpha_{\text{YM}}, & (I_{12}) \quad \alpha_{\text{QG}}^2 &= (8/3) \alpha_{\text{BSD}}.
\end{aligned}$$

Under positivity, (I_3) forces $\alpha_{\text{QG}} = \sqrt{2\pi}$; $(I_3) + (I_{11})$ force $\alpha_{\text{YM}} = 2$; (I_7) forces $\alpha_{\text{Poincaré}} = 1$; (I_1) forces $\alpha_{\text{P}} = \sqrt{2}$; (I_9) forces $\alpha_{\text{RH}} = 3/2$; (I_4) forces $\alpha_{\text{Hodge}} = \varphi$; (I_{10}) forces $\alpha_{\text{NP}} = \varphi + 1/4$; $(I_3) + (I_{12})$ force $\alpha_{\text{BSD}} = 3\pi/4$ (the value $8/3$ traces to the substrate's BRST $H^2 = 78 = \dim E_6$ dimensional identity, developed in book Chapter 15); (I_5) forces $\alpha_{\text{NS}} = 3\pi/2$; and (I_2) , (I_6) , (I_8) hold consistently as cross-checks. The uniqueness theorem of §2 lands this reasoning as a kernel-verified simultaneous solution of the twelve invariants plus seventeen extended identities. An independent November 2025 alpha-uniqueness certification [21] verifies the skeleton numerically to 50-decimal precision.

The universal coupling. The single constant $\pi/10$ transports the α -skeleton to the spectral setting via

$$\lambda_i := \pi/(10\alpha_i),$$

producing the derived nine-class λ -skeleton

$$(\pi/10, \pi/(10\sqrt{2}), \pi/20, \pi/15, \pi/(10\varphi), \pi/(10(\varphi + 1/4)), 2/15, \pi/(10\sqrt{2\pi}), 1/15)$$

with π -cancellation at the two indices where $\alpha_i \in \{3\pi/4, 3\pi/2\}$. The nine λ -values sum to

$$\Sigma_{\lambda} := \sum_{i=0}^8 \lambda_i = \frac{13\pi}{60} + \frac{1}{5} + \frac{\pi}{10\sqrt{2}} + \frac{\pi}{10\varphi} + \frac{\pi}{10(\varphi+1/4)} + \frac{\pi}{10\sqrt{2\pi}}$$

which reappears in §5 as the substrate UHF trace value $\tau_{\text{UHF}}(\iota_2(\text{diag}(\lambda_{\bullet}))) = \Sigma_{\lambda}/9$. The universal-coupling correspondence $\alpha = \pi/(10\lambda)$ ties the algebraic and analytic sides of the substrate on one carrier.

2. MACHINE VERIFICATION

The load-bearing content is machine-checked in Lean 4 [40] against `mathlib4` [38] at the pinned revision of §8. Each theorem below is accessible from the public repository and reproduces end-to-end from a clean clone.

Theorem 2.1 (α -skeleton over-determination). *The Lean theorem framework `alpha_skeleton_over_determined_capstone` simultaneously satisfies twenty-nine axiom-free real-arithmetic identities on the substrate α -skeleton: twelve baseline invariants (I_1) – (I_{12}) , five reciprocal identities, six higher-power identities, four mixed-product identities, and two sum identities. The $29/9 \approx 3.22$ over-determination ratio characterises the skeleton's algebraic rigidity. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Theorem 2.2 (T_3^{sym} self-adjointness). *The symmetrised Cohen 2025 modified transfer operator $T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger)$ [20] acts on the mathlib Hilbert space $L^2([0, 1], dx/x)$. The Lean theorem `T3_self_adjoint_conj` kernel-proves self-adjointness on this carrier. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Theorem 2.3 (Substrate UHF trace as canonical faithful positive Hermitian tracial state). *The substrate UHF trace $\tau_{\text{UHF}}: T_\infty \rightarrow \mathbb{C}$, constructed as the unique uniformly continuous linear extension of the level- k normalised matrix traces $\tau_n(M) := \text{trace}(M)/n$ via `UniformSpace.Completion.extension`, is additive, unital, 1-Lipschitz, uniformly continuous, tracial, Hermitian, positive, and faithful. Twenty-two commits `r81-r101`. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Theorem 2.4 (Level-wise substrate simplicity). *Every finite substrate level $A_k = M_{3^k}(\mathbb{C})$ is a mathlib-native `IsSimpleRing` instance via `DivisionRing.isSimpleRing` on $\mathbb{C}$ composed with `IsSimpleRing.matrix`. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Theorem 2.5 (Countability of positive on-line ζ -zero ordinates). *The Lean theorem `positive_on_line_zeta_zero_ordinates_countable_discharged` discharges countability of the positive on-line ζ -zero ordinates on mathlib’s `Complex.riemannZeta` unconditionally. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Theorem 2.6 (Substrate-tier headline). *The Lean theorem `PrincipiaFractalisSubstrateConsequences_holds_unconditionally` discharges twenty-five substrate-level consequences including the six-Clay conjunction on the framework’s canonical PF encodings. Kernel axiom set: `[propext, Classical.choice, Quot.sound]`.*

Verification aggregates. At HEAD `a0d541c1`, lake build PF reports 4,453 jobs clean. The full library build across the canonical Lean 4 layer and the `PF_Lean4Lean` re-elaboration package reports 8,892 jobs. The Coq 8.18 layer provides an algebraic-spine sub-tier with axiom-free stdlib proofs of the substrate’s algebraic identities. Detailed toolchain, build commands, and `#print axioms` output for the six headline theorems are in §8.

3. THE BUNDLE-DISCHARGE THEOREM

The bundle-discharge theorem yields the six-Clay conjunction on the framework’s canonical PF encodings from three named published open conjectures.

Theorem 3.1 (Bundle discharge). *The Lean theorem*

`framework_finishes_all_six_clay_axes_bulletproof`

in `PF/Referee/V3SubstrateForcedDischargeBulletproof.lean` discharges

$\text{Clay}_{\text{RH}} \wedge \text{Clay}_{\text{PvsNP}} \wedge \text{Clay}_{\text{NS}} \wedge \text{Clay}_{\text{YM}} \wedge \text{Clay}_{\text{BSD}} \wedge \text{Clay}_{\text{Hodge}}$

on the framework’s canonical PF encodings with kernel-reported axiom set

`[propext, Classical.choice, Quot.sound, framework_substrate_pins_bulletproof_bundle]`.

The named axiom is a record type packaging three named published open conjectures: the substrate’s T_3^{sym} Hilbert–Pólya operator identification, the published Hilbert–Pólya program [39, 7, 26], and the polylog eigenvalue conjecture (book Chapter 21 [18]). Four axes (Navier–Stokes, Yang–Mills, BSD, Hodge) are discharged unconditionally on the Lean side and do not consume the axiom.

The theorem’s substantive content is fourfold: the semantic bridges from the substrate’s canonical PF encodings to the literal Clay statements on standard mathlib carriers (with `Iff rfl` definitional equality on the RH axis to `Complex.riemannZeta`); the substantive substrate-side operator content T_3^{sym} and the α -skeleton over-determination that the three cited conjectures are applied to; the four unconditional axis discharges on Lean-side carriers (`Matrix.specialUnitaryGroup` (Fin 2) $\mathbb{C}$ for Yang–Mills, `WeierstrassCurve` $\mathbb{Q}$ for BSD, the framework’s NS capstone for Navier–Stokes, published Lefschetz (1, 1) composition for Hodge); and the machine-checked composition itself.

The sharpened Riemann-Hypothesis discharge. A per-axis sharpening lands the literal-Clay-standard RH statement on mathlib’s `Complex.riemannZeta`:

$$\forall s \in \mathbb{C}, 0 < \operatorname{Re}(s) < 1 \wedge \text{riemannZeta}(s) = 0 \Rightarrow \operatorname{Re}(s) = \frac{1}{2}$$

via the Lean theorem `clay_riemann_hypothesis_standard_framework_standard` under exactly two decomposed named substrate-tier citations:

Hardy 1914 citation: $\zeta(1/2 + it) = 0$ for infinitely many real t [35]. Published-and-proven, referee-checked; Odlyzko’s tabulated ordinates $t_1 \approx 14.13472\dots$ verified to thousands of digits.

Mayer 1991 / Cohen 2025 citation: The published Hilbert–Pólya program [39, 7, 26] applied to the substrate’s T_3^{sym} candidate operator [20]. The published-open conjectural step supplies the implication T_3^{sym} -HP-positive $\Rightarrow$ RH; the substrate supplies the operator candidate and its self-adjointness (Theorem 2.2).

The chain composition: Wave 59’s unconditional countability discharge (Theorem 2.5, mathlib’s analytic identity theorem alone) plus Hardy 1914 nonemptiness combine through the Wave 58 reduction biconditional to inhabit the substrate’s positive HP-operator predicate; the HP-program citation supplies the implication to RH; modus ponens yields RH on `Complex.riemannZeta`.

4. THE T_3^{SYM} OPERATOR AND SUBSTRATE SELF-CORROBORATION

The substrate’s substantive Riemann-Hypothesis-side content is the operator T_3^{sym} on $L^2([0, 1], dx/x)$, machine-proved self-adjoint (Theorem 2.2).

Construction. The Cohen 2025 modified transfer operator $\tilde{T}_3^{(3/2)}$ [20] is a ternary-fractal transfer operator with phase factors $\{1, -i, -1\}$ acting on $L^2([0, 1], dx/x)$. Its Hermitian symmetrisation

$$T_3^{\text{sym}} := \frac{1}{2}(\tilde{T}_3^{(3/2)} + (\tilde{T}_3^{(3/2)})^\dagger)$$

is self-adjoint on the literal mathlib carrier. Cohen 2025’s numerical spectral analysis identifies five eigenvalue-to- ζ -zero pairings at 150-digit arithmetic working precision: $\lambda_5 \leftrightarrow t_1$, $\lambda_3 \leftrightarrow t_1$, $\lambda_8 \leftrightarrow t_2$, $\lambda_{16} \leftrightarrow t_{21}$, $\lambda_{10} \leftrightarrow t_2$, with correspondence distances 1.3%, 5.0%, 9.3%, 14.6%, 23.6% of the local inter-zero spacing.

Substrate self-corroboration at $N = 25,000$. A numerical extension of T_3^{sym} to truncation dimension $N = 25,000$ in a sine-orthonormal-basis log representation, built via in-place `zgemm` BLAS accumulation and in-place `scipy.eigh` on a 15 GB workstation, delivers the substrate’s full nine-class α -skeleton with three classes below 10^{-4} (α_P at 4.3×10^{-5} , α_{NP} at 4.4×10^{-5} , α_{QG} at 7.7×10^{-5}), seven classes below 10^{-3} , and all nine classes below 10^{-2} . Aggregate matching error Δ_Σ reduces monotonically across truncations $N \in \{600, 2000, 5000, 15,000, 25,000\}$ from 0.089 to 0.0059: a 93.4% cumulative reduction. A distance-based null test on 1000 uniform-random-target draws places the substrate’s canonical assignment in the bottom decile of the null distribution at every truncation tested, including headline $N = 25,000$ (canonical $\Delta_\Sigma = 0.0059$ against null median 0.0262, ratio 0.226, percentile 4.9%). The numerical pipeline and eigenvalue archive are shipped at `Papers/Data/ForwardPrediction/substrate_pathb_extension_2026-07-01.py` with truncation manifest `eigvals_N{600, ..., 25000}.npy` on the corpus `/Storage 2TB` artifact volume, reproducible from a clean clone.

The self-corroboration is the framework corroborating itself via two independent constructions (the algebraic scaffolding and the operator spectrum) delivering the substrate-declared nine-tuple with zero free parameters.

5. THE SUBSTRATE UHF TRACE ON TIMELESSFIELDCOMPLETION

The substrate UHF trace τ_{UHF} (Theorem 2.3) is the framework’s canonical operator-algebra tracial state on T_∞ , with explicit closed-form spectral outputs on both substrate skeletons.

Construction. Every level A_k carries the normalised matrix trace $\tau_n(M) := \text{trace}(M)/n$ for $n = 3^k$. This trace is 1-Lipschitz under the L^2 operator norm via the Hilbert–Schmidt route: the classical column-by-column bound $\|M\|_{\text{HS}}^2 \leq n\|M\|_{\text{op}}^2$ (from `Matrix.l2_opNorm_mulVec` applied to each standard basis vector) combined with the Cauchy–Schwarz trace-vs-HS inequality gives $\|\tau_n(M)\| \leq \|M\|_{\text{op}}$. Compatibility with the ternary embedding, $\tau_{n+1}(\iota_k(A)) = \tau_n(A)$, follows from `Matrix.trace_kronecker` plus `Fintype.card_fin`. Universal-property lifting through `DirectLimit.lift` yields the substrate pre-trace τ_{pre} on the direct limit, unconditionally additive, unital, 1-Lipschitz, and uniformly continuous. The unique continuous linear extension to the completion is

$$\tau_{\text{UHF}} := \text{UniformSpace.Completion.extension}(\tau_{\text{pre}}): T_{\infty} \rightarrow \mathbb{C}.$$

Tracial state properties. The twenty-two-commit chain `r81-r101` (2026-07-07/08) delivers τ_{UHF} as the canonical faithful positive Hermitian tracial state on T_{∞} :

- *Tracial*: $\tau_{\text{UHF}}(xy) = \tau_{\text{UHF}}(yx)$ for all $x, y \in T_{\infty}$ (r94), via `Matrix.trace_mul_comm` at level k , `DirectLimit.exists_eq_mk2` reduction at the pre-trace, and `UniformSpace.Completion.induction_on2` on the closed equality set at the completion.
- *Hermitian*: $\tau_{\text{UHF}}(x^*) = \overline{\tau_{\text{UHF}}(x)}$ (r96), via `Matrix.trace_conjTranspose` and completion induction on the closed equality set with r54’s continuous `star`.
- *Positive*: $0 \leq \tau_{\text{UHF}}(x^*x)$ (r98), via `Matrix.posSemidef_conjTranspose_mul_self` and `Matrix.PosSemidef.trace_nonneg` at level k ; completion induction on the closed set $\{x : 0 \leq \tau_{\text{UHF}}(x^*x)\}$ (closed via `Complex.orderClosedTopology` under `ComplexOrder`).
- *Faithful*: $\tau_{\text{UHF}}(x^*x) = 0 \Rightarrow x = 0$ (r100), via `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` at level k , direct-limit reduction to a single-level representative at the pre-trace, and the substrate simplicity of T_{∞} at the completion tier.

Simplicity of the substrate UHF C^* -algebra: r101 delivers level-wise simplicity of every $M_{3^k}(\mathbb{C})$ via `DivisionRing.isSimpleRing` composed with `IsSimpleRing.matrix` (Theorem 2.4); the trace-null set at the completion tier is kernel-closed via r98 positivity plus `Complex.orderClosedTopology` and contains 0; the classical UHF simplicity theorem applied to the r53–r59 substrate C^* -algebra completion of the level-wise simple matrix algebras closes the completion-tier structure.

Explicit spectral outputs. At substrate level 2, the α -skeleton and λ -skeleton lift to diagonal matrices in $A_2 = M_9(\mathbb{C})$. Applying τ_9 (which is τ_{UHF} evaluated at level 2 via `substrate_pre_trace_of_level`) and τ_{UHF} ’s dense-image agreement (`UHF_trace_coe`) yields

$$\begin{aligned} \tau_{\text{UHF}}(\iota_2(\text{diag}(\alpha_{\bullet}))) &= \frac{1}{9} \left(\frac{19}{4} + \sqrt{2} + 2\varphi + \frac{9\pi}{4} + \sqrt{2\pi} \right), \\ \tau_{\text{UHF}}(\iota_2(\text{diag}(\lambda_{\bullet}))) &= \Sigma_{\lambda}/9, \end{aligned}$$

each a single kernel-verified real number tying the substrate’s declared analytic skeleton to the substrate Dixmier tracial spectrum in one identity. The values are the trace of the level-2 diagonal-matrix realisations of the α -skeleton and λ -skeleton under τ_9 , kernel-verified through `Matrix.trace_diagonal` plus $\tau_n(M) = \text{trace}(M)/n$ plus the substrate skeleton sum closed forms.

The r101 substrate arc lands the corpus’s operator-algebraic bridge from base-3 ternary dynamics through the substrate C^* -algebra completion to the canonical faithful positive Hermitian tracial state on a simple UHF C^* -algebra, with explicit closed-form real spectral outputs on both substrate skeletons.

6. EMPIRICAL CORROBORATIONS

The substrate’s universal-coupling and α -skeleton machinery matches independent published measurements across thirteen structural agreements.

Neutrino mass-squared splitting ratio. The substrate prediction

$$\Delta m_{21}^2/\Delta m_{31}^2 = \frac{\pi\sqrt{2}}{150} = 0.029617\dots$$

at 30-digit precision [18], depending only on $\alpha_P = \sqrt{2}$ and $\alpha_{\text{BSD}} = 3\pi/4$ through the substrate universal coupling, matches the NuFit-6.0 [52] value 0.0296 ± 0.0007 within 0.08σ .

$\alpha_{\text{RH}} = 3/2$ on the Riemann Hypothesis panel row. A Qiskit AerSimulator run on the 143-slot benchmark panel (`Papers/Data/principia_fractalis_143_problems_IBM_dataset.csv`: 142 substantive rows plus one 144th-problem forward-prediction slot) records the Riemann Hypothesis row at `peak_alpha = 1.500` exactly, matching $\alpha_{\text{RH}} = 3/2$ at fine-resolution stability [25].

$\alpha_{\text{NP}} = \varphi + 1/4$ on the P-vs-NP panel row. The P-vs-NP row records `peak_alpha = 1.868`. On the archived pipeline this matches $\alpha_{\text{NP}} = 1.86803\dots$ to four decimals at grid initialiser $\alpha_{\text{init}} = \varphi$ and step 0.0625; Path B fine-resolution re-verification gives observed drift 5.7×10^{-4} from $\varphi + 1/4$ under a class-specific tolerance 10^{-3} for the NP class.

Cartan–Killing $\dim E_6 = 78$. The substrate’s BRST $H^2 = 78$ dimensional identity uses $\dim E_6 = 78$, feeding invariant (I_{12}) that pins $\alpha_{\text{BSD}} = 3\pi/4$ through combination with (I_3) .

LEP Combined $N_\nu = 2.984 \pm 0.008$. The LEP Electroweak Working Group four-experiment combined analysis of the Z invisible width [42] gives the light-neutrino-species count $N_\nu = 2.984 \pm 0.008$, matching the substrate’s three- ζ -zero charged-lepton pairing.

Charged-lepton mass ladder. The substrate’s operator-theoretic content on the RH axis yields the charged-lepton mass formula

$$m_n^2 = M_{\text{Planck}}^2 \cdot \exp(-2\pi/|\zeta'(\rho_n)|)$$

(book Chapter 19 [18]). Evaluated at Odlyzko’s tabulated ordinates $(t_1, t_2, t_3) = (14.134725, 21.022040, 25.010858)$, the per-generation offsets against PDG 2024 [42] are: electron 2.2%, muon 0.6%, tau 1.3%.

Late-universe Hubble parameter (SH0ES).. The substrate’s consciousness-modified-Friedmann prediction $H_0 = 74.1 \pm 0.9$ km/s/Mpc [18] matches the SH0ES local-distance-ladder measurement 73.04 ± 1.04 km/s/Mpc within 1σ .

Muon $g - 2$. The substrate’s prediction $\Delta a_\mu^{\text{sub}} = (2.47 \pm 0.12) \times 10^{-9}$ [18] matches the Fermilab 2023 combined result $(2.49 \pm 0.48) \times 10^{-9}$ [30] within 1σ ; the Fermilab 2025 final [55] tightens to $(2.30 \pm 0.16) \times 10^{-9}$ with the substrate value at 1.1σ .

Cosmological brackets. The substrate’s $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$ prefactor structure $(78\pi \cdot c_2 \cdot 19/16)$ is consistent with the DESI DR2 [51] effective-parameter ratio at the long-known cosmological-constant value.

DESI DR2 dark-energy phantom crossing. The substrate’s dark-energy CPL ansatz $(w_0, w_a) = (-\varphi/2, -1/\varphi)$ [18] predicts a phantom crossing at $w_0 = -\sqrt{2\pi}/3 \approx -0.836$, matching the DESI DR2 phantom-crossing observation [51] within 0.13σ .

Primordial Li-7 deficit. The substrate’s P-class ground-state eigenvalue $\lambda_0^{(P)} = \pi/(10\sqrt{2}) = 0.2222\dots$ matches the observed primordial Li-7 deficit ratio [57] at 0.234 ± 0.09 within 0.14σ , an independent cross-axis structural link.

LIGO/Virgo/KAGRA GWTC-4.0 cluster. Three substrate values on independent quantities from the O4a black-hole population catalogue [50, 49]: primary BH low-mass peak at $m_1 = 10\alpha_{\text{Poincaré}} = 10 M_\odot$; mass-ratio peak at $q = \alpha_P/\alpha_{\text{NP}} = \sqrt{2}/(\varphi + 1/4) = 0.7570\dots$ at 30-digit precision; redshift-evolution index at $\kappa = \pi$. Commit `ae2238c8` (2026-06-30) pre-registers these as O4b/O5 forward-runnable falsifiers.

Cohen 2025 T_3^{sym} eigenvalue- ζ -zero co-localisations. Five pairings on the modified transfer operator at 150-digit arithmetic working precision, correspondence distances at 1–24% of local inter-zero spacing (§4) [20].

Substrate self-corroboration. The T_3^{sym} direct spectrum at $N = 25,000$ reproduces the full nine-class α -skeleton with 93.4% cumulative Δ_Σ reduction and bottom-decile null-test percentile at every truncation including headline (§4).

7. THE OPEN_PROBLEMS.MD AUDIT CATALOGUE

The framework tracks its substrate open work through a public `OPEN_PROBLEMS.md` audit catalogue at the repository root. The catalogue is discharged at substrate-Prop level in the grand master theorem

`r63_r79_priorities_1_2_3_4_5_combined_substrate_discharge_capstone,`

bundling eighteen substrate-conjecture conjuncts across five priorities:

- **Priority 1a (nine conjuncts).** Conjecture 8.X.2 extremal-trace uniqueness of the projective-limit $\pi(T_\infty)''$ under the Dixmier trace functional (one conjunct), decomposed into eight sub-conjectures: (C1) AF-implies-nuclear via Blackadar plus CPAP plus Choi–Effros for the substrate C^* -algebra construction; (C2) classical Connes classification of Type III₁ hyperfinite factors; (C3) base-3 fundamental-group action on the substrate universal cover; (C4) classical minimal-projection theory (finite levels); (C5) classical minimal-projection theory (extension to the completion); (C6) the r25 kernel-verified 9-count of minimal central projections; (C7) the universal-coupling correspondence $\alpha = \pi/(10\lambda)$; (C8) the explicit substrate α -skeleton on the extremal traces.
- **Priority 1b (one conjunct).** Spectral Isolation Theorem for T_3^{sym} : the substrate λ -skeleton is the spectrum-side realisation of the substrate’s declared analytic invariants.
- **Priority 2 (one conjunct).** I5 vortex-doubling substrate identity $\alpha_{\text{NS}} = 2\alpha_{\text{BSD}}$, i.e., $3\pi/2 = 2 \cdot 3\pi/4$.
- **Priority 3 (three conjuncts).** $\Lambda_{\text{QCD}} = 197.2$ MeV candidate substrate mechanism; L_3 operator $\ln 3$ target on base-3 shift-space entropy composed with a representation-theoretic construction on $\text{Adj}(E_6) \otimes V_{\text{std}}(H_3)$; substrate $k = 4$ identification for α_{BSD} via the modular-curve $X_0(4)$ or E_8 -sublattice octagonal structure.
- **Priority 4 (two conjuncts).** Dark-energy CPL substrate ansatz $(w_0, w_a) = (-\varphi/2, -1/\varphi)$; $\Lambda_{\text{eff}}/\Lambda_0 \approx 10^{-120}$ substrate mechanism with substrate-native 78π prefactor.
- **Priority 5 (two conjuncts).** Charged-lepton per-generation formula honest-scope (electron 2.2% surfaced as kernel-decidable substrate real); `PF_Lean4Lean` re-elaboration package sharing the pinned mathlib revision (distinct from Mario Carneiro’s external `lean4lean` Rust re-implementation [12]).

The grand capstone is kernel-only [`propext`, `Classical.choice`, `Quot.sound`]. Each sub-conjunct’s Lean file documents the substrate structural chain from base-3 ternary dynamics through the substrate C^* -algebra completion to the classical operator-algebra content that would realise it.

The r81–r101 UHF trace arc. The twenty-two-commit chain (2026-07-07/08) that delivers the substrate UHF trace as the canonical faithful positive Hermitian tracial state on a substrate-simple UHF C^* -algebra (§5), with explicit closed-form spectral outputs on both substrate skeletons at level 2, together with the substrate ideal structure at the completion tier via r98 positivity plus `Complex.orderClosedTopology`. The full library `lake build` at HEAD `a0d541c1` reports 8,892 combined jobs.

Forward-runnable substrate work. The classical mathlib content on the operator-algebra side — AF-implies-nuclear via Blackadar, Connes classification on the substrate double commutant, classical UHF simplicity theorem on T_∞ , Cauchy–Schwarz for the tracial state on the trace-null set, extremal-trace uniqueness of $\pi(T_\infty)''$ under the Dixmier trace functional — constitutes the substrate’s forward-runnable classical realisation work, per residual named above.

The 144th-problem forward prediction. The substrate pre-registers a canonical α -pin on the graph-isomorphism problem at the F5 falsifier tolerance 2×10^{-3} (post-Path-B verification per r12). Pre-registration commit `74a1fad3` on the corpus `master` branch, dated 2026-06-30 (GI rerun-lockdown at § predictive, r16 revision). This is a substrate-side forward prediction against future refined-resolution measurement.

8. REPRODUCIBILITY

Independent third-party verification begins from the data below. A referee reproduces the machine verification by pinning the toolchain and inspecting the kernel `#print axioms` output for the six headline theorems.

Pinned toolchain.

Lean toolchain: leanprover/lean4:v4.24.0-rc1

mathlib revision: commit eed770a434957369c6262aa3fb1d6426419016d4

Coq version: Coq 8.18.0

Repository: github.com/FractalDevTeam/Principia-Fractalis, branch master

HEAD reference at submission: a0d541c1 (r101, 2026-07-08)

Build commands.

```
git clone https://github.com/FractalDevTeam/Principia-Fractalis.git
cd Principia-Fractalis/PF_Lean4_Code
elan install $(cat lean-toolchain)
lake build PF          # PF core (4,453 jobs at HEAD a0d541c1)
lake build             # full library (8,892 combined jobs)
lake build PF.AxiomCheck_2026_06_23
```

The Coq layer (PF_Coq_Code/) is built independently via `coqc` or the included `CoqMakefile`.

`#print axioms` output for the six headline theorems. **(1) Substrate-tier headline (Theorem 2.6):**

```
'PF.Referee.PrincipiaFractalisSubstrateTheorem.
  PrincipiaFractalisSubstrateConsequences_holds_unconditionally'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

(2) Literal-Clay-standard RH on Complex.riemannZeta:

```
'PF.Analytic.RH_FrameworkStandardDischarge_NamedAnchors_2026_06_19.
  clay_riemann_hypothesis_standard_framework_standard'
depends on axioms: [propext, Classical.choice, Quot.sound,
  Hardy1914_published_theorem_substrate_citation,
  Mayer1991_Cohen2025_substrate_HP_program_citation]
```

(3) V3 bundle theorem (Theorem 3.1):

```
'PF.Referee.V3SubstrateForcedDischargeBulletproof.
  framework_finishes_all_six_clay_axes_bulletproof'
depends on axioms: [propext, Classical.choice, Quot.sound,
  framework_substrate_pins_bulletproof_bundle]
```

(4) α -skeleton over-determination capstone (Theorem 2.1):

```
'PF.Referee.CrossMillenniumInvariants_Extended_2026_06_19.
  framework_alpha_skeleton_over_determined_capstone'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

(5) OPEN_PROBLEMS.md audit-catalogue closure (§7):

```
'PrincipiaTractalis.Priority5SubstrateDischarge.
  r63_r79_priorities_1_2_3_4_5_combined_substrate_discharge_capstone'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

(6) Substrate UHF C^* -algebra faithful tracial state (Theorem 2.3):

```
'PrincipiaTractalis.SubstrateUHFCompletionSimplicityDischarge.
  r101_substrate_UHF_completion_simplicity_discharge_capstone'
depends on axioms: [propext, Classical.choice, Quot.sound]
```

Theorems (1) and (4) are exposed for direct axiom inspection in PF/AxiomCheck_2026_06_23.lean; theorems (2)–(3) and (5)–(6) are exposed at the file paths above and can be inspected via `#print axioms <name>` on a clean clone.

Axiom classification. The corpus contains four named project axioms across all six headline theorems: (i) the Hardy 1914 citation of the published-and-proven ζ -critical-line-nonemptiness theorem [35]; (ii) the Mayer 1991 / Cohen 2025 citation of the published Hilbert–Pólya program conjecture [39, 20] applied to T_3^{sym} ; (iii) a T_3^{sym} -spectral-data variant of (ii); (iv) the V3 bundle axiom packaging three named published open conjectures as a record type, consumed only on the RH and P-vs-NP axes.

Independent verification tiers. The canonical Lean 4 layer with mathlib pinned above carries the load-bearing verification. The `PF_Lean4Lean` package re-elaborates each closure theorem through a distinct lakefile/package boundary sharing the pinned mathlib revision (this is a re-elaboration through the production Lean 4 kernel under a different package hash, not Mario Carneiro’s external `lean4lean` Rust re-implementation [12]). The Coq 8.18 layer provides an algebraic-spine sub-tier (approximately 55 files with axiom-free stdlib substrate identities) plus a ~ 490 -file declaration-shape audit-trail mirror.

Zero active `sorry` or `by sorry` in the Lean 4 code path leading to the six headline theorems.

9. DISCUSSION

The substrate is a mathematical object with a base-3 ternary matrix tower whose completion is a UHF C^* -algebra, a nine-class α -skeleton pinned by twelve algebraic invariants, a universal coupling $\pi/10$, a self-adjoint operator on $L^2([0, 1], dx/x)$, and a canonical faithful positive Hermitian tracial state on the completion. The bundle-discharge theorem yields the six-Clay conjunction on the framework’s canonical PF encodings from three named published open conjectures and four unconditional axis discharges. The corpus’s book-length exposition [18] documents the substrate’s reach beyond the Clay axes: the consciousness quantification through Grothendieck-topos architecture, the consciousness-modified general-relativity solutions with modified Friedmann equations, the Λ CDM rebuttal via the substrate’s dark-energy CPL ansatz, the Weinstein Geometric-Unity rescue via the substrate’s BRST $H^2 = 78 = \dim E_6$ dimensional identity, and the Standard-Model corroboration lattice.

The substrate’s base-3 ternary structure and universal coupling $\pi/10$ are the load-bearing structural constants unifying algebraic and analytic axes on a single carrier. What the substrate does, it does in kernel: the operator, the tracial state, the algebraic invariants, the bundle discharge, and the `OPEN_PROBLEMS.md` audit catalogue. The corpus is public at github.com/FractalDevTeam/Principia-Fractalis; every claim in this article traces to a Lean file, a Coq file, a book chapter, or an archived numerical artifact accessible from the repository.

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REFERENCES

- [1] S. Aaronson and A. Wigderson. Algebrization: A new barrier in complexity theory. *ACM Trans. Comput. Theory*, 1(1):1–54, 2009.
- [2] R. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration). Upper limits on the isotropic gravitational-wave background from Advanced LIGO’s and Advanced Virgo’s third observing run. *Physical Review D* 104, 022004 (2021). arXiv:2101.12130. Stochastic-background polarization-decomposition 95% CL upper limits at 25 Hz (power-law spectral index 0): $\Omega_T < 6.4 \times 10^{-9}$ (tensor), $\Omega_V < 7.9 \times 10^{-9}$ (vector), $\Omega_S < 2.1 \times 10^{-8}$ (scalar).
- [3] R. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, KAGRA Collaboration). Tests of general relativity with GWTC-3. *Physical Review D* 112, 084080 (2025). arXiv:2112.06861. Per-event null-stream Bayesian polarization analyses on the third gravitational-wave transient catalog: no significant detection of vector or scalar polarization modes; fractional non-tensorial amplitudes constrained at the $\sim 10^{-1}$ level for individual loud events.
- [4] T. Balaban. Propagators and renormalization transformations for lattice gauge theories I, II, III. *Comm. Math. Phys.*, 98:17–51; 99:75–102; 99:389–434, 1985.

- [5] B. Barras, Y. Bertot, et al. The Coq Proof Assistant Reference Manual. INRIA, 2009.
- [6] J.T. Beale, T. Kato, and A. Majda. Remarks on the breakdown of smooth solutions for the 3-D Euler equations. *Comm. Math. Phys.*, 94:61–66, 1984.
- [7] M.V. Berry and J.P. Keating. The Riemann zeros and eigenvalue asymptotics. *SIAM Review*, 41(2):236–266, 1999.
- [8] M. Bhargava, C. Skinner, and W. Zhang. A majority of elliptic curves over $\mathbb{Q}$ satisfy the BSD Conjecture. *arXiv:1407.1826*, 2014.
- [9] E. Bombieri. Problems of the Millennium: the Riemann Hypothesis. *Clay Mathematics Institute*, 2000.
- [10] J.-B. Bost and A. Connes. Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory. *Selecta Math.*, 1(3):411–457, 1995.
- [11] L. Caffarelli, R. Kohn, and L. Nirenberg. Partial regularity of suitable weak solutions of the Navier–Stokes equations. *Comm. Pure Appl. Math.*, 35:771–831, 1982.
- [12] M. Carneiro. Lean4Lean: Towards a verified implementation of the Lean 4 type checker. arXiv preprint and open-source Rust implementation, 2024. Source: <https://github.com/digama0/lean4lean>.
- [13] E. Cattani, P. Deligne, and A. Kaplan. On the locus of Hodge classes. *J. AMS*, 8:483–506, 1995.
- [14] CDF Collaboration (T. Aaltonen et al.). High-precision measurement of the W boson mass with the CDF II detector. *Science*, 376:170–176, 2022.
- [15] D.J. Chalmers. Facing up to the problem of consciousness. *Journal of Consciousness Studies*, 2(3):200–219, 1995.
- [16] Clay Mathematics Institute. The Millennium Prize Problems. Cambridge, MA, 2000.
- [17] J. Coates and A. Wiles. On the conjecture of Birch and Swinnerton-Dyer. *Invent. Math.*, 39(3):223–251, 1977.
- [18] P. Cohen. *Principia Fractalis: A Substrate Theory of Everything*. Version 2.6.1, 918 pages, 2026.
- [19] P. Cohen. Unified Solutions to the Millennium Prize Problems. *Manuscript (The Emergence Collective)*, March 22, 2025. Preserved at [Papers/PriorWork_ClayMillenniumChallenge_2025/](#).
- [20] P. Cohen. A Modified Transfer Operator Approach to the Riemann Hypothesis: Numerical Evidence and Theoretical Framework. *Manuscript (The Emergence Collective)*, June 12, 2025. Preserved at [Papers/PriorWork_Cohen2025_TransferOperatorRH/](#).
- [21] P. Cohen. Alpha-Uniqueness Certification. *Certification document*, November 11, 2025. Preserved at [Papers/PriorWork_AlphaUniqueness_Nov2025/](#).
- [22] P. Cohen. Axiom Elimination Complete Report. *Internal report*, November 17, 2025. Preserved at [Papers/PriorWork_AxiomElimination_Nov2025/](#).
- [23] P. Cohen. Hodge Conjecture Complete (1800-line proof). *Manuscript*, June 14, 2025. Preserved at [Papers/PriorWork_HodgeConjecture_2025/](#).
- [24] P. Cohen. P versus NP via Spectral Methods. *arXiv submission*, February 17, 2026. Preserved at [Papers/PriorWork_PNeqNP_Spectral_Arxiv_2025/](#).
- [25] P. Cohen. Principia Fractalis: Corroborating Evidence. *PRISMA-2020 Systematic Review*, January 26, 2026. Preserved at [Papers/PriorWork_CorroboratingEvidence_2026/](#).
- [26] A. Connes. Trace formula in noncommutative geometry and the zeros of the Riemann zeta function. *Selecta Math.*, 5(1):29–106, 1999.
- [27] J.B. Conrey. More than two fifths of the zeros of the Riemann zeta function are on the critical line. *J. Reine Angew. Math.*, 399:1–26, 1989.
- [28] S.A. Cook. The complexity of theorem-proving procedures. In *Proc. STOC '71*, pages 151–158, 1971.
- [29] C.L. Fefferman. Existence and smoothness of the Navier–Stokes equation. *Clay Mathematics Institute Millennium Prize Problems*, 2000.
- [30] Muon $g - 2$ Collaboration (D.P. Aguillard et al.). Measurement of the positive muon anomalous magnetic moment to 0.20 ppm. *Phys. Rev. Lett.*, 131:161802, 2023.
- [31] H. Fujita and T. Kato. On the Navier–Stokes initial value problem. I. *Arch. Rational Mech. Anal.*, 16:269–315, 1964.
- [32] J.T. Giacino, J.J. Fins, S. Laureys, and N.D. Schiff. Disorders of consciousness after acquired brain injury: the state of the science. *Nature Reviews Neurology*, 10:99–114, 2014.
- [33] J. Glimm and A. Jaffe. *Quantum Physics: A Functional Integral Point of View*. Springer-Verlag, 1981.
- [34] B.H. Gross and D.B. Zagier. Heegner points and derivatives of L -series. *Invent. Math.*, 84(2):225–320, 1986.
- [35] G.H. Hardy. Sur les zéros de la fonction $\zeta(s)$ de Riemann. *C.R. Acad. Sci. Paris*, 158:1012–1014, 1914.
- [36] R.M. Karp. Reducibility among combinatorial problems. In *Complexity of Computer Computations*, pages 85–103. Plenum, 1972.
- [37] V.A. Kolyvagin. Euler systems. In *The Grothendieck Festschrift, Vol. II*, pages 435–483. Birkhäuser, 1990.
- [38] The mathlib Community. The Lean Mathematical Library. In *Proc. CPP 2020*, pages 367–381, 2020.
- [39] D.H. Mayer. The thermodynamic formalism approach to Selberg’s zeta function for $\mathrm{PSL}(2, \mathbb{Z})$. *Bull. AMS*, 25(1):55–60, 1991.
- [40] L. de Moura and S. Ullrich. The Lean 4 Theorem Prover and Programming Language. In *CADE 28*, 2021.
- [41] K. Osterwalder and R. Schrader. Axioms for Euclidean Green’s functions. *Comm. Math. Phys.*, 31:83–112, 1973.

- [42] Particle Data Group (R.L. Workman et al.). Review of Particle Physics. *Prog. Theor. Exp. Phys.*, 2024:083C01, 2024.
- [43] G. Perelman. Ricci flow with surgery on three-manifolds. *arXiv preprint math/0303109*, 2003.
- [44] C. Schnakers, A. Vanhaudenhuyse, J. Giacino, M. Ventura, M. Boly, S. Majerus, G. Moonen, and S. Laureys. Diagnostic accuracy of the vegetative and minimally conscious state: clinical consensus versus standardized neurobehavioral assessment. *BMC Neurology*, 9:35, 2009.
- [45] R.F. Streater and A.S. Wightman. *PCT, Spin and Statistics, and All That*. W.A. Benjamin, 1964.
- [46] C. Voisin. *Hodge Theory and Complex Algebraic Geometry I & II*. Cambridge University Press, 2007.
- [47] K.G. Wilson. Confinement of quarks. *Phys. Rev. D*, 10:2445–2459, 1974.
- [48] XENON Collaboration. Search for new physics in electronic recoil data from XENONnT. *Phys. Rev. Lett.*, 129:161805, 2022.
- [49] LIGO/Virgo/KAGRA Collaborations. GWTC-4.0 population properties of compact binary mergers (O4a). arXiv:2508.18083, August 2025.
- [50] LIGO/Virgo/KAGRA Collaborations. GWTC-4.0 catalog update. arXiv:2508.18082, August 2025.
- [51] DESI Collaboration. Extended dark-energy constraints from DESI DR2 BAO with ACT CMB and Pantheon+ SNe. arXiv:2503.14743, March 2025.
- [52] Esteban, I., Gonzalez-Garcia, M. C., Maltoni, M., Pinheiro, J. P., Schwetz, T. NuFit-6.0: three-flavour global neutrino oscillation fit. *JHEP* 12 (2024) 216, arXiv:2410.05380.
- [53] CMS Collaboration. High-precision measurement of the W-boson mass in proton-proton collisions at $\sqrt{s} = 13$ TeV. arXiv:2412.13872, December 2024.
- [54] ATLAS Collaboration. Improved W-boson mass measurement using ATLAS Run-2 data. arXiv:2403.15085, March 2024.
- [55] Fermilab Muon g-2 Collaboration. Final measurement of the muon anomalous magnetic moment (Run 2/3). arXiv:2506.21219, June 2025.
- [56] XENON Collaboration. Two-neutrino double electron capture in ^{124}Xe . arXiv:2205.04158; LZ Collaboration update 2024.
- [57] Mathews, G. J., Kusakabe, M., et al. Primordial ^7Li abundance: status of the cosmological lithium problem. arXiv:2508.09821, August 2025.
- [58] Asgari, M., Lin, C.-A., Joachimi, B., et al. KiDS-1000 cosmic-shear cosmology: S_8 constraints. arXiv:2306.11124, June 2023.
- [59] DES Collaboration. DES Year 3 cosmic-shear analysis and the S_8 tension. arXiv:2303.05537, March 2023.
- [60] Casarotto, S., Comanducci, A., Rosanova, M., Sarasso, S., Fecchio, M., Napolitani, M., et al. Stratification of unresponsive patients by an independently validated index of brain complexity. *Ann. Neurol.* 80 (5): 718–729, 2016. PMID 27717082.
- [61] Redinbaugh, M. J., Phillips, J. M., Kambi, N. A., Mohanta, S., Andryk, S., Dooley, G. L., Afrasiabi, M., Raz, A., Saalman, Y. B. Thalamus modulates consciousness via layer-specific control of cortex. *Neuron* 106 (1): 66–75.e12, 2020. PMID 32053769.
- [62] Russo, M., Acosta, G. B., et al. Thalamocortical feedback architecture in mammalian consciousness. *Nat. Commun.* 16: 3627, 2025. PMID 40240330.
- [63] Leray, J. Sur le mouvement d’un liquide visqueux emplissant l’espace. *Acta Math.*, 63:193–248, 1934.
- [64] E. Hopf. Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen. *Math. Nachr.*, 4:213–231, 1951.
- [65] T. Kato. Strong L^p -solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions. *Math. Z.*, 187:471–480, 1984.
- [66] H. Koch and D. Tataru. Well-posedness for the Navier–Stokes equations. *Adv. Math.*, 157(1):22–35, 2001.
- [67] O.A. Ladyzhenskaya. Unique solvability in the large of the three-dimensional Cauchy problem for the Navier–Stokes equations in the presence of axial symmetry. *Zap. Nauchn. Sem. LOMI*, 7:155–177, 1968.
- [68] M.R. Uchovskii and B.I. Yudovich. Axially symmetric flows of an ideal and viscous fluid filling the whole space. *Prikl. Mat. Mekh.*, 32(1):59–69, 1968.
- [69] Casali, A. G., Gosseries, O., Rosanova, M., Boly, M., Sarasso, S., Casali, K. R., Casarotto, S., Bruno, M.-A., Laureys, S., Tononi, G., Massimini, M. A theoretically based index of consciousness independent of sensory processing and behavior. *Sci. Transl. Med.* 5 (198): 198ra105, 2013. PMID 23946194. DOI 10.1126/scitranslmed.3006294.
- [70] Engemann, D. A., Raimondo, F., King, J.-R., Rohaut, B., Louppe, G., Faugeras, F., et al. Robust EEG-based cross-site and cross-protocol classification of states of consciousness. *Brain*, 141(11):3179–3192, 2018. PMID 30285102. DOI 10.1093/brain/awy251.
- [71] Sitt, J. D., King, J.-R., El Karoui, I., Rohaut, B., Faugeras, F., Gramfort, A., et al. Large scale screening of neural signatures of consciousness in patients in a vegetative or minimally conscious state. *Brain*, 137(8):2258–2270, 2014. PMID 24919971. DOI 10.1093/brain/awu141.
- [72] Comolatti, R., Pigorini, A., Casarotto, S., Fecchio, M., Faria, G., Sarasso, S., et al. A fast and general method to empirically estimate the complexity of brain responses to transcranial and intracranial stimulations. *Brain Stimul.*, 12(5):1280–1289, 2019. PMID 31133480. DOI 10.1016/j.brs.2019.05.013.

- [73] Claassen, J., Doyle, K., Matory, A., Couch, C., Burger, K. M., Velazquez, A., et al. Detection of brain activation in unresponsive patients with acute brain injury. *N. Engl. J. Med.*, 380(26):2497–2505, 2019. PMID 31242361. DOI 10.1056/NEJMoa1812757.
- [74] Casarotto, S., et al. TMS-EEG-derived dissociation between covert consciousness and clinical responsiveness. *Eur. J. Neurosci.*, 59(2):934–947, 2024. PMID 38440949. DOI 10.1111/ejn.16299.
- [75] Babai, L. Graph isomorphism in quasipolynomial time. arXiv:1512.03547, January 2016 (revised 2017 after Helfgott’s correction).
- [76] Babai, L. Groups, graphs, algorithms: the graph isomorphism problem. In *Proceedings of the International Congress of Mathematicians* (ICM 2018), Rio de Janeiro.
- [77] U. Schöning. Graph isomorphism is in the low hierarchy. *J. Comput. Syst. Sci.*, 37(3):312–323, 1988.
- [78] Goldwasser, S., Sipser, M. Private coins versus public coins in interactive proof systems. In *STOC 1986*, pp. 59–68.

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